Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form		
Lab Name	**Implementing Different Network Topologies in Cisco Packet Tracer**	
Lab Number	**03**	
Submitted by:		
Student Names	**Registration #**	
Sumaira Safeer	FA22-BCE-019	

Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab Tasks

Statement Purpose:

Get acquainted with Packet Tracer. And Make some simple Packet Tracer scenarios.

Activity outcomes:

- Students will have gained the basic understanding of Packet Tracer to see "protocols in action".
- Students will have developed basic understanding of digging deep into the network protocols.

Introduction to the Packet Tracer Interface.

Network Topology:

Network topology is the way devices are connected in a network. It defines how these components are connected and how data transfer between the network. There are two major categories of Network Topology i.e. Physical Network topology and Logical Network Topology.

Types of Network Topology:

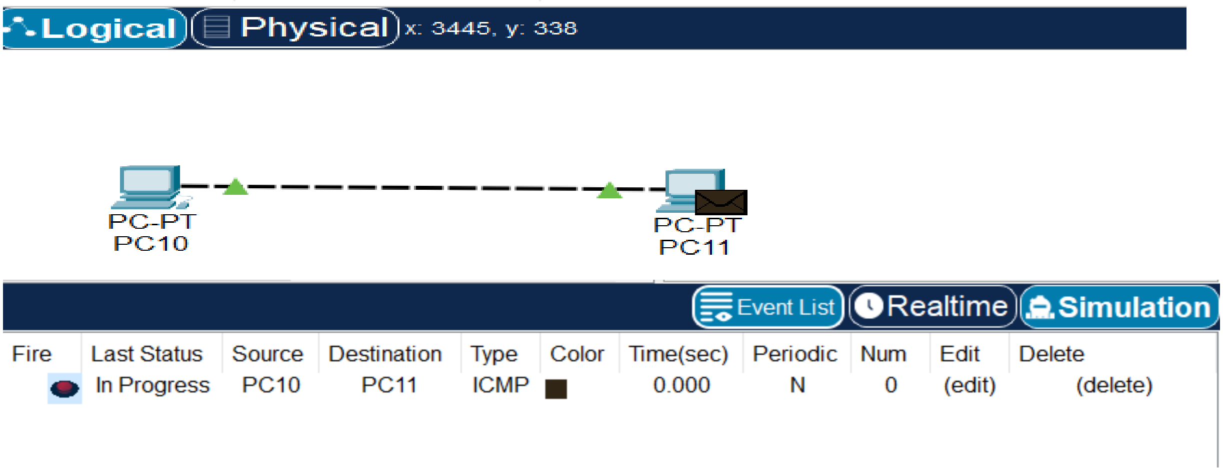
Below mentioned are the types of Network Topology.

- Point to Point Topology
- Mesh Topology.
- Star Topology.
- Bus Topology.
- Ring Topology.
- Tree Topology.
- Hybrid Topology.

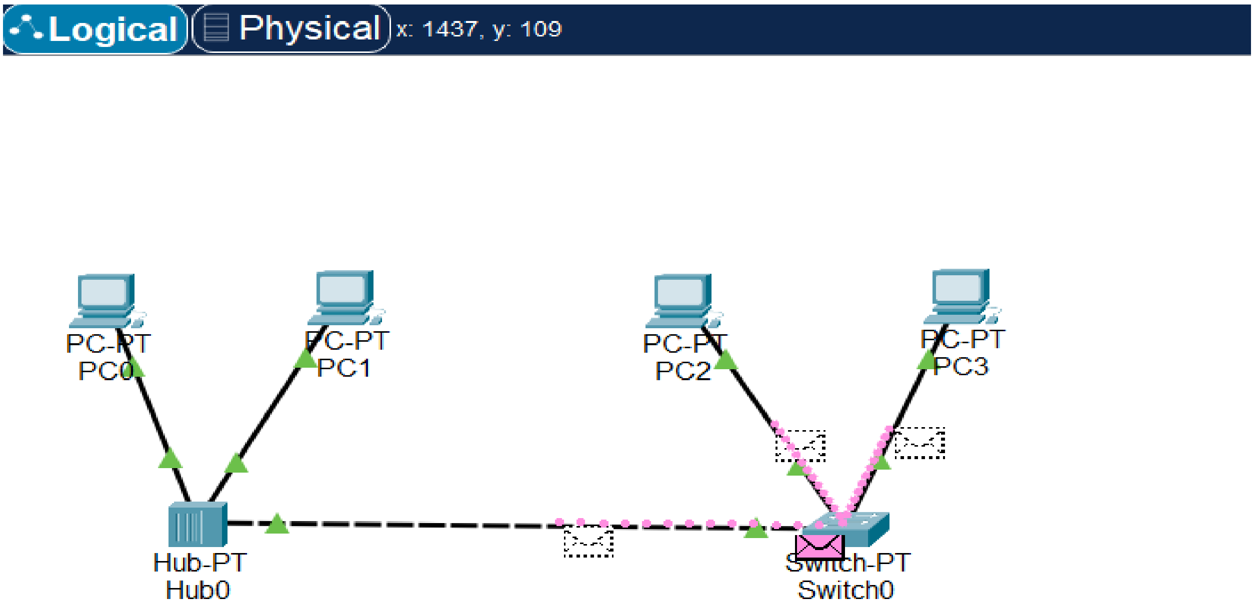
Activity 1:

Make topologies in packet tracer and provide connectivity:

1) **Point-to-Point:**

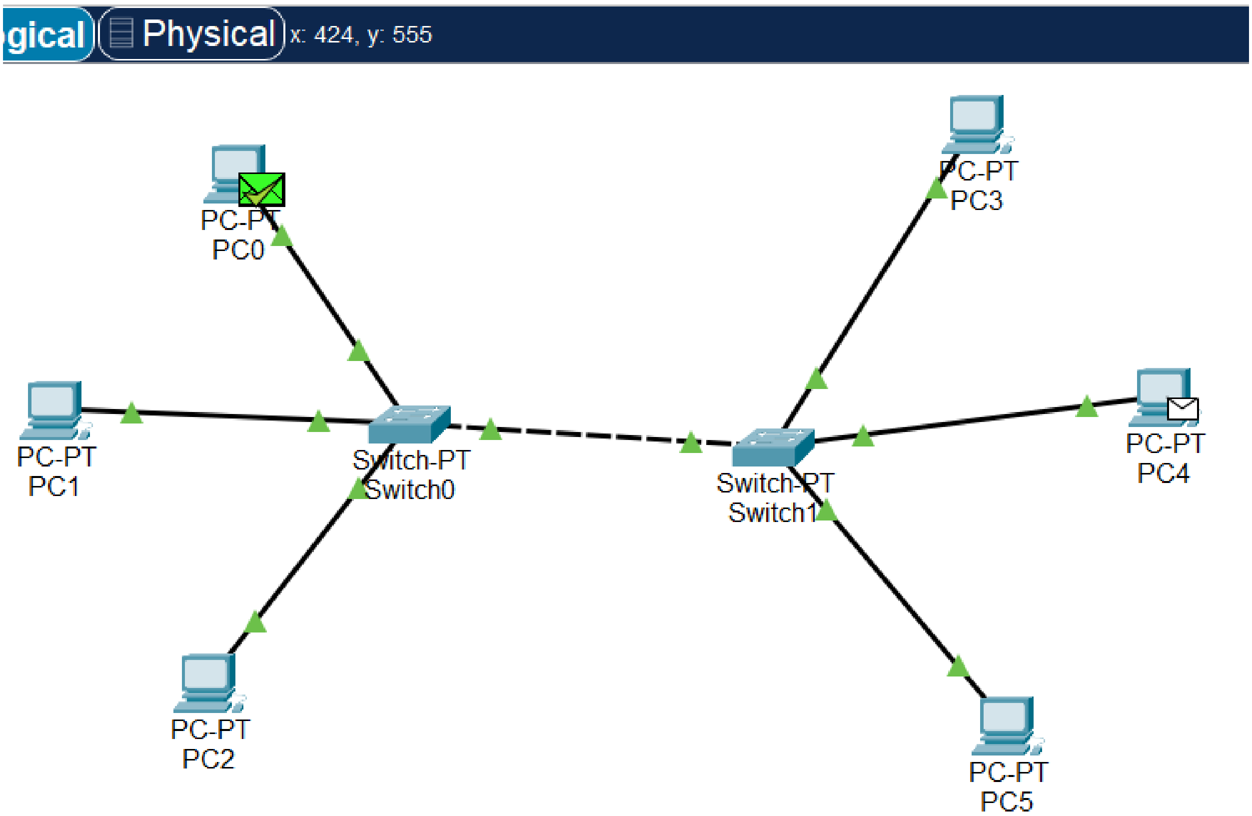


2) **Linear Bus:**



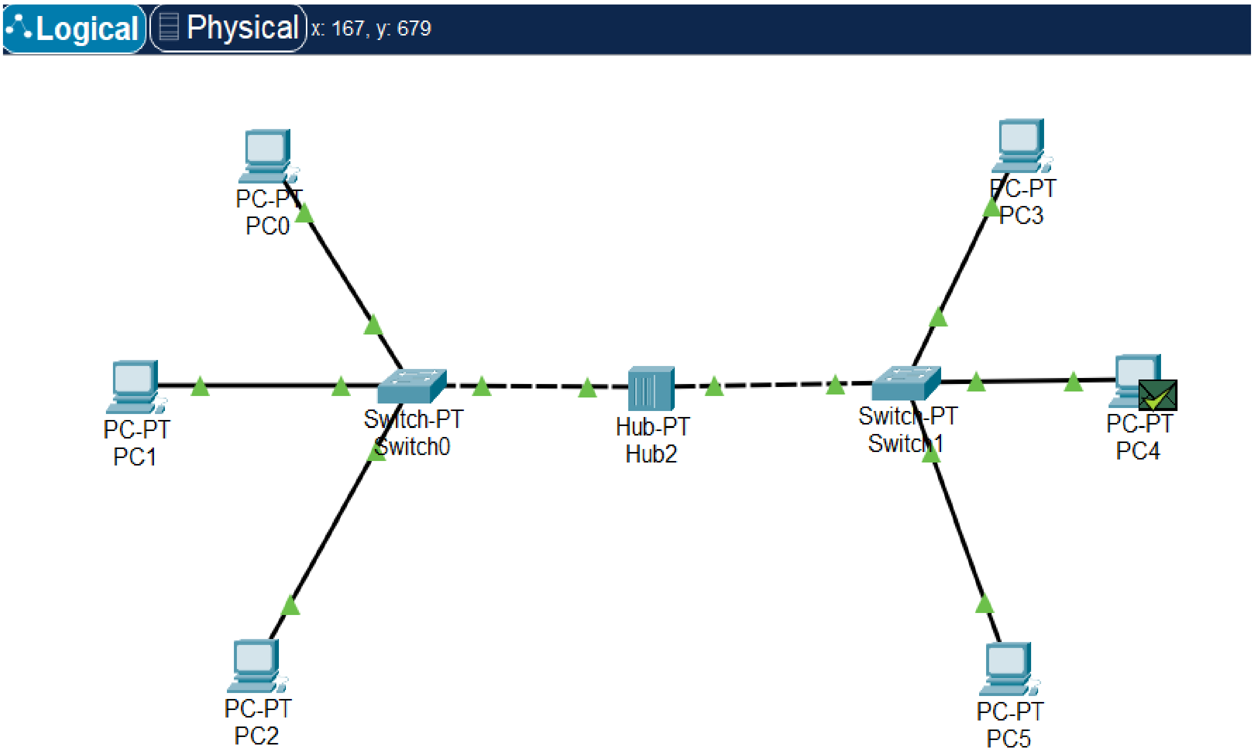
3) **Star Topology:**

a. **Extended star:**

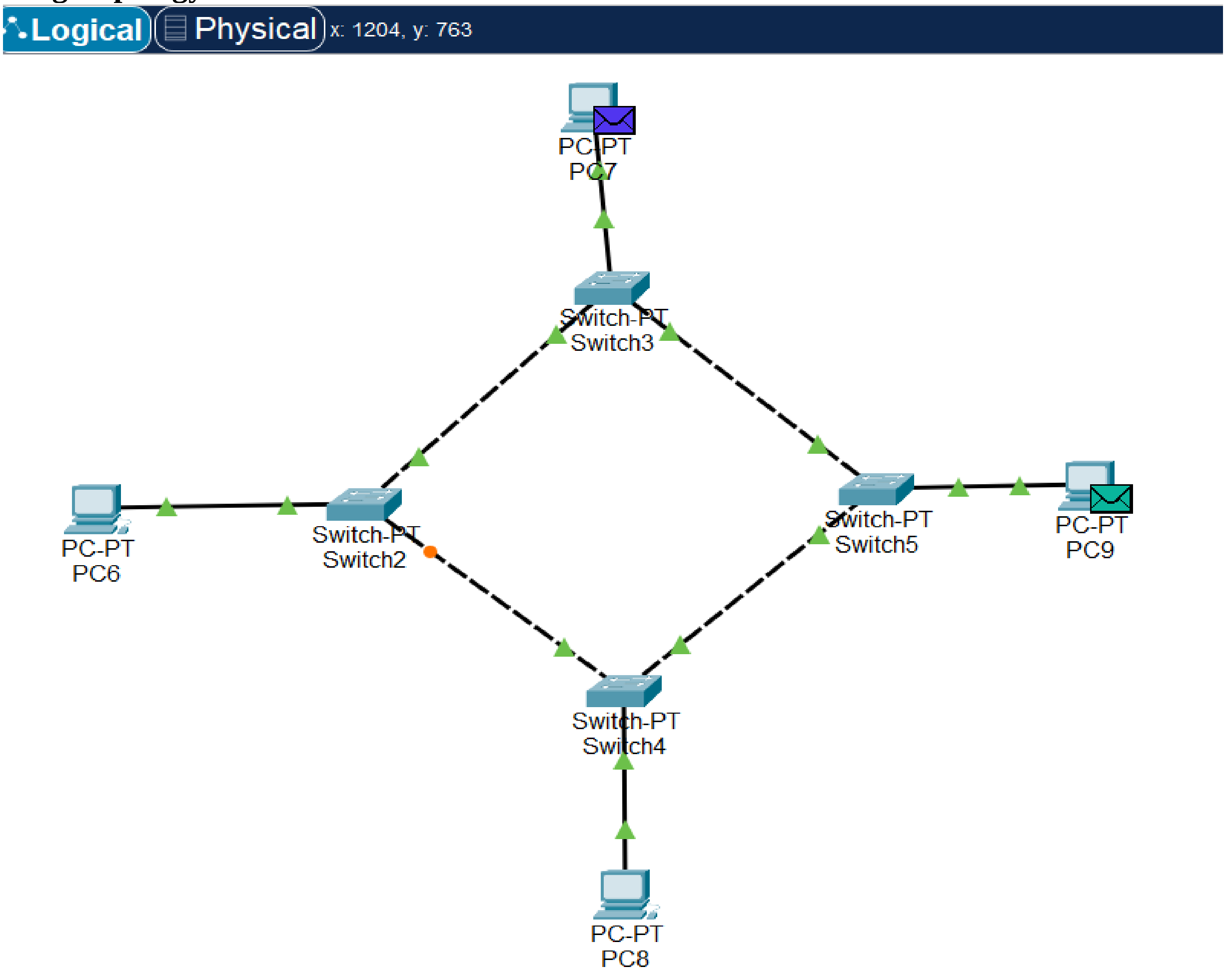


Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC5	PC4	ICMP		0.029	N	1	(edit)	(delete)

b. Distributed Star:

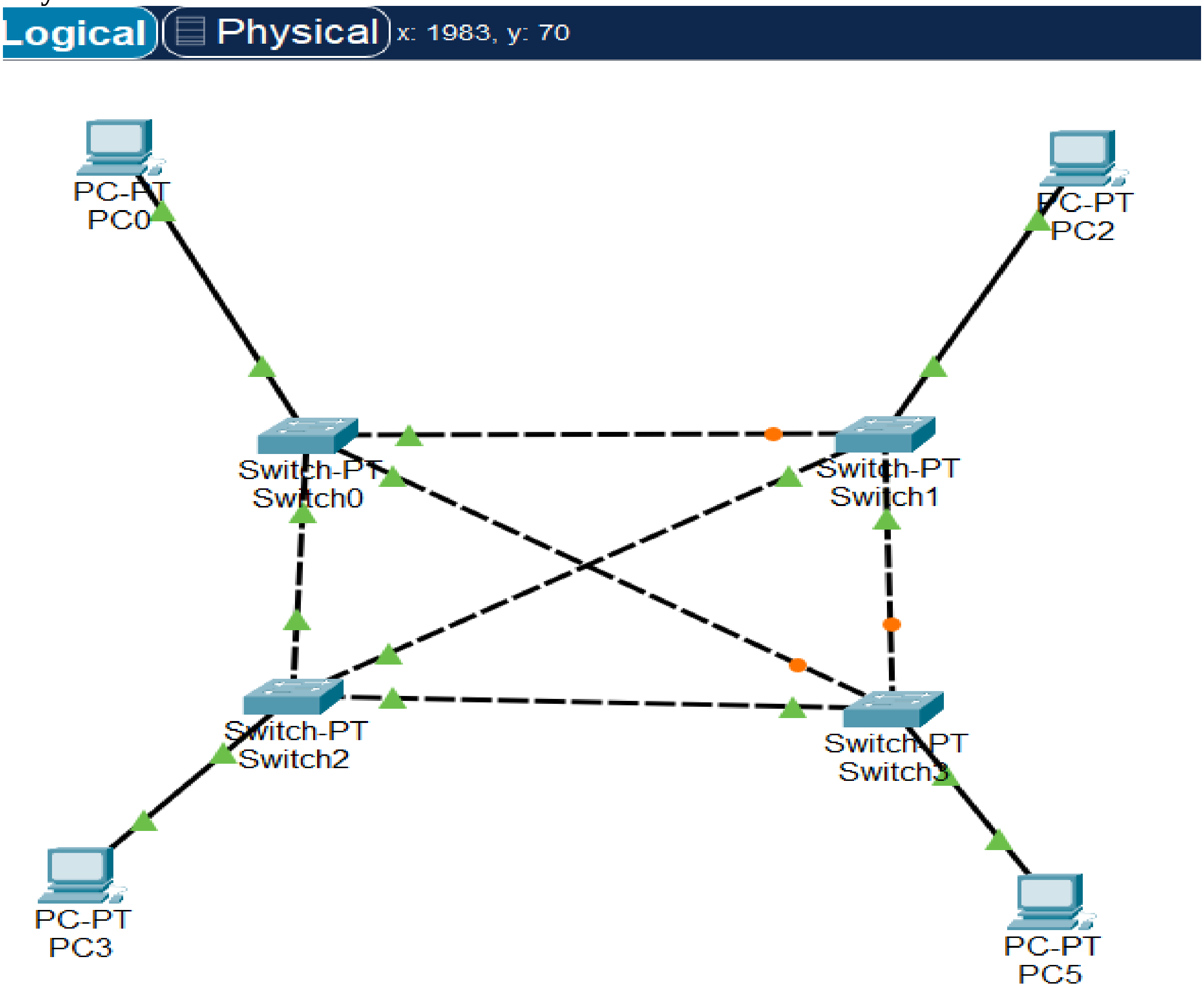


4) Ring Topology:

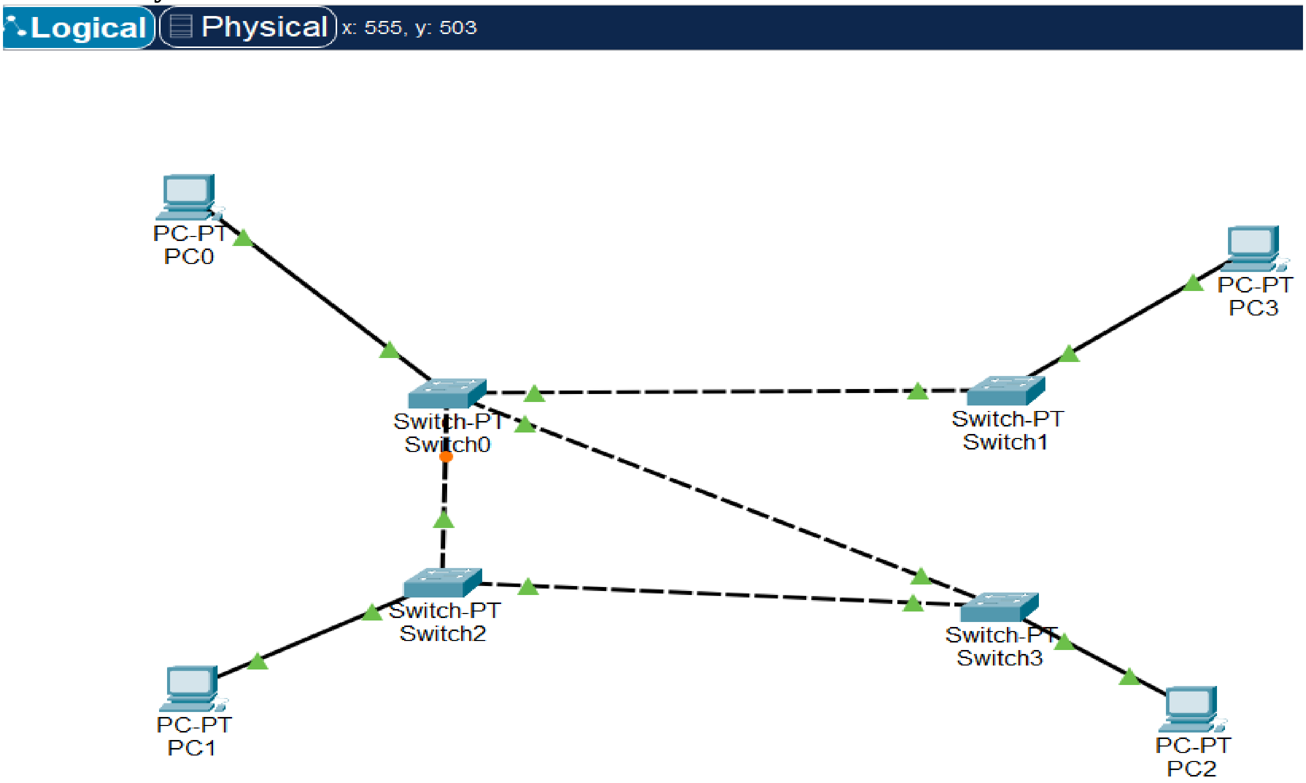


5) Mesh Topology:

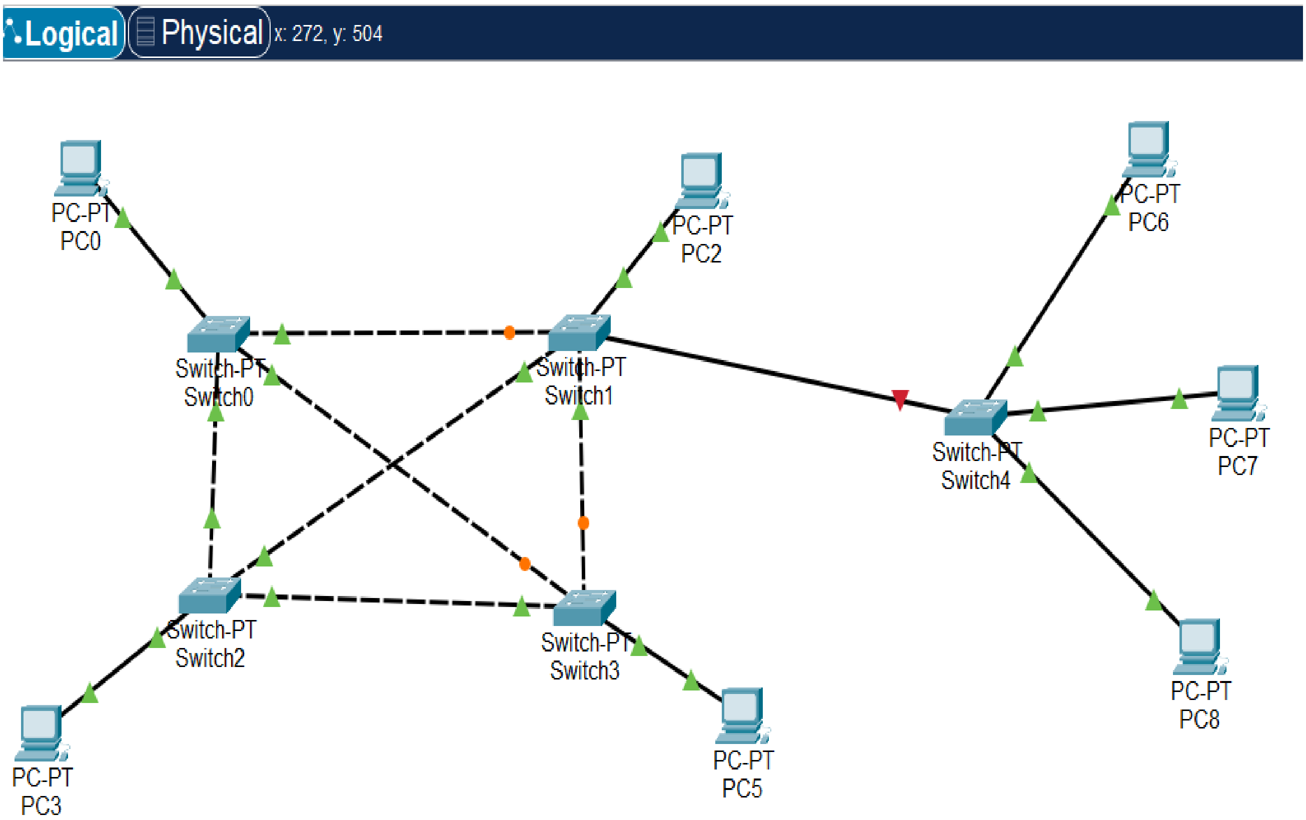
a. Fully connected network:



c. **Partially connected network:**



5) Hybrid Topology:



Conclusion:

In conclusion, network topologies play a crucial role in determining the efficiency and reliability of a computer network. Each topology, whether it's bus, star, ring, mesh, or tree, offers unique benefits and potential drawbacks. By understanding these different arrangements, network designers can choose the most appropriate topology to meet the specific needs of their systems, ensuring optimal performance and connectivity.